

Similarly, 95 percent of returns will fall within two standard deviations, and 99.7 percent within three standard deviations of the mean. So, if ABC’s mean return was 10 percent and its standard deviation was 15 percent, then two-thirds of all returns produced by ABC would equal 10 percent plus-or-minus no more than 15 percent; that is, 67 percent of the time, ABC’s return will likely be somewhere between –5 percent and 25 percent. It follows that 95 percent of ABC’s returns will fall between –20 percent and 40 percent and that 99.7 percent will fall somewhere between –35 percent and 55 percent.

Now, imagine that one, but not both, of the following questions is to appear on an investor’s risk tolerance questionnaire. Both concern Portfolio ABC, and both try to measure an investor’s comfort level with ABC, given its average returns, volatility, and so on. However, the two questions frame the situation very differently. As you compare questions 1 and 2, try to imagine how an average investor, probably subject to a few common behavioral biases, might respond to each respective frame. Do you think most investors’ answers would be identical in each instance?

1. Based on Table 12.1, which investment portfolio seems like the best fit, bearing in mind your own risk tolerance as well as your desire for long-term return?
- a. Portfolio XYZ

b. Portfolio DEF

c. Portfolio ABC
2. Assume that you own Portfolio ABC and that it lost 15 percent of its value over the past year, despite previous years of good performance. This loss is consistent with the performance of similar funds during the past year. What is your reaction to this situation?
- a. Sell all Portfolio ABC shares.

b. Sell some, but not all, Portfolio ABC shares.

c. Continue to hold Portfolio ABC shares.

d. Increase investment in Portfolio ABC.

TABLE 12.1 Portfolio Selection: Which Portfolio Seems Best?

Portfolio Number	95% Probability Gain/Loss Range	Long-Term Return
XYZ	2% to 4%	3%
DEF	–6% to 18%	6%
ABC	–20% to 40%	10%